International Journal of Educational Technology in Higher Education

RESEARCH ARTICLE Open Access

Intelligent teaching analytics for collaborative reflection: investigating pre-service teachers' perceptions, experiences and shared regulation processes

Mengke Wang[1,2], Zengzhao Chen[1*], Ying Xu[1], Bhagya Maheshi[2] and Dragan Gašević[2*]

*Correspondence:
llczz@163.com;
dragan.gasevic@monash.edu

[1] Faculty of Artificial Intelligence in Education, Central China Normal University, Wuhan, China
[2] Faculty of Information Technology, Monash University, Melbourne, Australia

Abstract

Reflection is important for pre-service teacher development as it can help deepen their understanding of teaching and benefit their teaching practice. There have been several proposals of intelligent teaching analytics developed with the aim to strengthen the connection between practice and reflection by providing feedback to pre-service teachers. However, the implementation and user experience of intelligent teaching analytics in collaborative reflection remain underexplored. This study aimed to explore pre-service teachers' experiences and shared regulation during collaborative reflection. Using a design-based research approach, 20 pre-service teachers participated in a seven-week collaborative reflection program. The intelligent teaching analytics implementation was iteratively improved over time, with data collected from reflection processes and interviews. Analysis of shared regulation revealed that as the implementation was refined over time, the interconnections among different regulation behaviors increased. Thematic analysis further indicated that pre-service teachers perceived intelligent teaching analytics as beneficial for enhancing their pedagogical content knowledge. Emotionally, they initially felt nervous and embarrassed but gradually found the collaborative reflection atmosphere to be friendly, harmonious, and mutually supportive. This research contributes to the development of principles for integrating intelligent teaching analytics into collaborative reflection and provide suggestions for future advancement of intelligent teaching analytics.

Keywords: AI in teacher education, Collaborative reflection, Feedback, Intelligent teaching analytics, Shared regulation

Introduction

The rapid integration of artificial intelligence (AI) into education has driven global efforts to empower teachers by providing them with the tools and insights needed to excel in the digital age (Ndukwe & Daniel, 2020; UNESCO, 2024). A key way AI supports teachers is by analyzing teaching processes and providing feedback to enhance their reflective practices. Teaching is inherently dynamic and complex, requiring educators to engage learners, foster knowledge construction, and address diverse student

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

needs. However, pre-service teachers often face significant barriers in this process. Due to their limited professional vision, they may struggle to fully identify teaching issues or recognize opportunities for improvement (Prilop et al., 2021). This is where intelligent teaching analytics can play a critical role, providing data-driven insights to support their reflection.

Intelligent teaching analytics, which refers to the use of AI technologies to automate and optimize the analysis of teaching practices, aims to enhance teaching quality through diagnostic and analytical approaches (Ndukwe & Daniel, 2020). The application of intelligent teaching analytics in teaching reflection is believed to support informed decision-making and facilitate evidence-based practice (Lee et al., 2024). Recent research has developed theoretical models and frameworks designed to leverage intelligent teaching analytics for enhancing teacher reflection. For instance, Ndukwe and Daniel (2020) introduced the Teaching Outcome Model (TOM), a data-driven framework that utilizes teaching analytics to guide actions for improving teaching performance. Similarly, Qazi and Pachler et al. (2024) proposed a conceptual framework for integrating data analytics into teacher professional learning, leveraging digital technologies to provide personalized, context-aware professional development recommendations. In addition, several studies have explored the application of intelligent teaching analytics in teacher reflection. For instance, Jacobs et al. (2022) investigated how primary and secondary school teachers used the TalkMoves app to receive feedback on 'Accountable Talk', examining its potential impact on teacher reflection and teaching practice. Cai et al. (2024) conducted an experimental study showing that an AI-based dashboard—a data summary tool utilizing intelligent speech recognition and natural language processing to analyze classroom interactions—significantly improved the quality of teachers' reflections compared to a control group without the AI tool.

However, research on the effective implementation and user experiences of intelligent teaching analytics in collaborative reflection remains limited (Bond, 2024). Collaborative reflection, defined as a process of collaborative critical thinking involving cognitive and affective interactions among individuals to achieve new intersubjective understandings (Huang et al., 2021), is widely recognized as a key component of effective teacher learning (Kelly, 2006). Although some studies, such as Cai et al. (2024), have applied intelligent teaching analytics in collaborative reflection, their focus has been on evaluating its effects compared to non-usage, rather than examining implementation processes. Additionally, little research has examined users' perceptions and experiences across different stages of intelligent teaching analytics implementation. These aspects are crucial for understanding the effectiveness and impact of intelligent teaching analytics in supporting collaborative reflection. Furthermore, as a group activity**,** collaborative reflection inherently involves shared regulation—the coordination of metacognitive, emotional, and motivational processes among participants—which plays a vital role in shaping the depth of reflection and enhancing learning outcomes (Järvelä et al., 2016; Ucan & Webb, 2015). Examining shared regulation can offers insights into the core mechanisms of collaborative reflection.

Therefore, this study aimed to explore the application of intelligent teaching analytics to promote collaborative reflection, as well as examine pre-service teachers' perception, experience and shared regulation during its implementation. Based on the findings,

this paper discusses implications for refining intelligent teaching analytics to better meet pre-service teachers' reflection needs, along with principles for incorporating intelligent teaching analytics into collaborative reflection.

Literature review

The potential and challenges of intelligent teaching analytics in promoting teaching reflection

The concept of teaching analytics is central to the growing data-rich technology-enhanced learning and teaching environment (van Leeuwen et al., 2022). Teaching analytics is closely related to learning analytics (Gašević et al., 2015), but with a distinct focus. While learning analytics concentrates on analyzing learners' learning processes and outcomes through data, teaching analytics specifically aims to assess and improve teaching practices, ultimately enhancing teaching effectiveness and supporting informed instructional decisions. Intelligent teaching analytics emphasize the technological foundation of AI technologies such as natural language processing and deep machine learning (Poonja et al., 2023; Sümer et al., 2021) and advanced visualization techniques such as data storytelling (Echeverria et al., 2024). The results of intelligent teaching analytics are typically presented through data-driven, visual dashboards that help teachers reflect on and make informed decisions (van Leeuwen et al., 2022). By analyzing the teaching process and providing feedback to teachers, intelligent teaching analytics can bridge the gap between teaching practice and reflection.

The potential of intelligent teaching analytics in promoting teaching reflection

Teaching reflection is informed by the evaluation of teaching performance. Traditional teaching evaluation relies on manual observation, whereas intelligent teaching analytics provides several distinct advantages compared to manual observation. First, it can analyze behaviors that are labor-intensive for human observers and exhibit specific patterns, such as estimating teachers' eye gaze direction (Shuo et al., 2022),thus complementing manual observation and enabling more comprehensive feedback (Wise & Jung, 2019). Second, intelligent teaching analytics can efficiently process large volumes of data, identify patterns and trends, and deliver timely, continuous feedback to facilitate teaching reflection (Martinez-Maldonado et al., 2022; Ndukwe & Daniel, 2020). Third, AI-driven algorithms are not affected by fatigue or emotional fluctuations, ensuring consistent results when analyzing the same teaching video at different times, and thus enhancing the reliability of outcomes. The above features align with the characteristics of effective feedback, which emphasize comprehensive, evidence-based, timeliness, and consistency (Johnson, 2016). In summary, intelligent teaching analytics promote reflection by offering new information and data-driven insights that enhance teachers' understanding of their practices and support continuous improvement.

The challenge of intelligent teaching analytics in promoting teaching reflection

Although intelligent teaching analytics offer significant benefits, they also face notable challenges. From a technical perspective, the complexity and opacity of algorithms may hinder pre-service teachers' understanding (Burrell, 2016; Khosravi et al., 2022). Moreover, trust issues caused by the "black box" nature of AI, further complicate adoption

(Kizilcec, 2016; Wang et al., 2025). In practice, models trained in controlled settings often struggle to perform effectively authentic and dynamic classroom environments (Martinez-Maldonado et al., 2023; Yoon et al., 2024). Additionally, teachers' limited data literacy, which refers to the ability to comprehend, analyze and interpret data and data visualization (Shreiner & Dykes, 2021), can result in data misinterpretation, reducing the utility of analytics for decision-making (Ndukwe & Daniel, 2020; Pozdniakov et al., 2023; van Geel et al., 2017). In summary, trust issues, data misinterpretation, and difficulties in data-informed decision-making, may hinder the effectiveness of intelligent teaching analytics in fostering reflection.

The application of intelligent teaching analytics in teaching reflection

Most existing studies on intelligent teaching analytics have focused on the developing and optimizing of AI platforms, such as Edusense (Ahuja et al., 2019), Moodoo (Martinez-Maldonado et al., 2022), and ACORN (Ramakrishnan et al., 2021). Recent research has increasingly paid attention to the practical application of intelligent teaching analytics. For example, Wang et al., (2014a, 2014b) used the Language ENvironment Analysis (LENA) tool in daily instructional practices to provide feedback on teacher-student dialogues, demonstrating its potential to enhance classroom interactions. However, challenges persist in strategically applying intelligent teaching analytics to teachers' reflective practices. For instance, Jacobs et al. (2022) examined the impact of TalkMoves, an automated discourse analysis tool, on teaching. Their findings indicate that teachers require additional support to effectively link conversational feedback to their teaching practice. Moreover, because of the noisy classroom environment, teachers expressed concerns about the tool's accuracy in capturing and transcribing teacher-student dialogues.

Therefore, research is lacking on effective approaches to integrating intelligent teaching analytics into teaching reflection, especially in a collaborative reflection setting. While the advantages and limitations have been broadly discussed, the specific impacts of applying intelligent teaching analytics with particular functionalities in collaborative reflection, as well as pre-service teachers' perceptions and experiences, remain unexplored.

The cognitive and emotional aspects in pre-service teachers' collaborative reflection

Collaborative reflection refers to the process in which a group of individuals, such as teachers, engage in reflective practice to analyze and improve their professional activities (Segal, 2024). It highlights the shift from viewing reflection as an individual cognitive process to understanding it as a social practice. As emphasized by socio-cultural theory, social interactions play a crucial role in facilitating professional learning (Vygotsky & Cole, 1978). Research on the effectiveness of collaborative reflection has yielded mixed results, with some studies highlighting its benefits in fostering deeper reflection and knowledge construction (Sorensen, 2014), while others report issues like poor levels of problem solving and perspective shift (Killeavy & Moloney, 2010; Tillema & Van der Westhuizen, 2006). Clarà et al.(2019) claim that the success of collaborative reflection largely depends on support from knowledgeable others, which can be facilitated by intelligent teaching analytics and the social organization of the collaboration.

The cognitive aspect in pre-service teachers' collaborative reflection

The cognitive aspect of collaborative reflection encompasses process such as task understanding, problem-solving, and knowledge construction . Reflection becomes more impactful when it extends beyond individual introspection and is incorporated into a community setting, building upon sociocultural frameworks (Jay & Johnson, 2002). However, several studies have identified significant cognitive challenges in pre-service teachers' collaborative reflection. Cognitive interactions during reflection depend on individual knowledge levels, with a higher degree of Technological Pedagogical Content Knowledge (TPACK) facilitating better cognitive engagement and more effective reflective thinking. Prior research highlights that while pre-service teachers often demonstrate strong Content Knowledge (CK), their Technological Pedagogical Knowledge (TPK) and TPACK tend to be weaker (Hao, 2016). Additionally, collaborative settings require pre-service teachers to balance personal perspectives with group dynamics, introducing additional cognitive and emotional demands (Segal, 2024). Moreover, as a collective activity, collaborative reflection risks becoming superficial if not carefully structured. Studies have shown that overly ritualized reflection, where participants have limited autonomy in determining when and how to contribute, diminished the depth and overall effectiveness of the reflective process (Ross, 2011).

To enhance the collaborative process, various cognitive tools have been developed, including visualization tools (Celepkolu et al., 2021), scripted collaboration tools (Miller & Hadwin, 2015), and technology-supported collaborative platforms (Unal & Cakir, 2021).With rapid advances in AI, intelligent teaching analytics tools have emerged as cutting-edge cognitive tools. Unlike earlier tools, intelligent teaching analytics has the potential to function as a "knowledgeable others" by providing data-driven insights and structured feedback to support the evaluation and improvement of teaching practices. Yet, integrating intelligent teaching analytics into pre-service teachers' collaborative reflection to deepen its effectiveness remains a critical area for future research.

The emotional aspect in pre-service teachers' collaborative reflection

Emotions are pervasive in collaborative reflection. From the perspective of emotional labor theory, teaching inherently involves emotional labor, and revisiting these emotional experiences during reflection can lead to emotional exhaustion (Tsang, 2011). It is widely accepted that cognitive dissonance, which happens when pre-service teacher discover discrepancies between their teaching behaviors, beliefs, or outcomes and their expectation drives reflection and change, but may also evoke emotions such as anxiety or frustration (King, 2014; Treacy & Leavy, 2023). Additionally, teacher self-efficacy research highlights that the practicum is a particularly stressful stage, during which teachers' emotional and physiological states fluctuate, significantly influencing their confidence and perceived competence (Bandura, 1997; Pfitzner-Eden, 2016). In collaborative reflection settings, the emotions of individual pre-service teachers can spread to other group members (Barsade et al., 2018), thereby affecting the overall atmosphere of the collaborative reflection. As a result, how group members engage in shared emotional regulation is crucial for maintaining positive emotional interactions (Huang & Lajoie, 2023). In summary, the collaborative reflection process is inherently emotional.

The interaction of cognitive and emotion aspect in collaborative reflection

In collaborative reflection, cognition and emotion are deeply interconnected, with emotional states influencing cognitive processes and vice versa. For example, positive emotions such as confidence or interest can enhance cognitive engagement and self-regulation, fostering deeper reflection and learning (Pekrun, 2006). Similarly, cognitive growth can improve emotional regulation; as individuals engage in meaningful reflection, their emotions may transition from frustration to satisfaction (De France & Hollenstein, 2022). Additionally, emotional support from peers, such as encouragement and validation, can help mitigate negative emotions, facilitating cognitive shifts and fostering positive changes in perspective (McConchie et al., 2022). However, this interplay is not without challenges. Emotional overload or negative group dynamics can hinder reflection, highlighting the need for structured support and a collaborative environment that fosters positive interactions.

Therefore, addressing both cognitive processes and social dynamics is essential for enhancing pre-service teachers' collaborative reflection. Creating a safe and trusting environment is crucial for fostering meaningful and effective reflective dialogue. Additionally, exploring how intelligent teaching analytics and emotional factors, such as the overall atmosphere, influence reflective interactions is a crucial area for future research.

The present study

The current study aimed to explore how intelligent teaching analytics could be effectively applied to pre-service teachers' collaborative reflection and to examine its impact. Based on this purpose, three research questions were formulated.

First, drawing on previous studies on technology acceptance, research suggests that users' perceptions of a tool influence both their willingness to use it and its effectiveness (Khechine et al., 2020; Wang et al., 2025). As intelligent teaching analytics are used to analyze teaching process, pre-service teachers' perceptions of it are crucial for its effective application in collaborative reflection. Moreover, understanding pre-service teachers' suggestions for improving the use of intelligent teaching analytics is equally important, as this can offer practical insights for future implementation. Based on these considerations, our first research question was:

RQ1: What are pre-service teachers' perceptions and suggestions for intelligent teaching analytics?

Second, existing literature suggests that cognitive and emotional experiences are interconnected and constitute an integral part of the learning process (Tan & Jung, 2024). During collaborative reflection, pre-service teachers inevitably undergo both cognitive and emotional experiences. Examining these experiences can provide valuable insights into the effectiveness of intelligent teaching analytics in supporting collaborative reflection. For instance, if pre-service teachers perceive intelligent teaching analytics as cognitively beneficial and report positive emotional experiences, it can be inferred that the application of intelligent teaching analytics effectively fosters collaborative reflection. Therefore, investigating pre-service teachers' cognitive and emotional experiences can provide a better understanding of the effectiveness of intelligent teaching analytics in collaborative reflection. Thus, our second research question was:

RQ2: What are pre-service teachers' cognitive and emotional experiences during intelligent teaching analytics-supported collaborative reflection?

Third, shared regulation is a group-level phenomenon manifested in collaborative learning, where participants iteratively negotiate and refine their understanding of the process while jointly managing cognitive, behavioral, motivational, and emotional aspects (Hadwin et al., 2017). As a strategic activity, it plays a crucial role in shaping participation dynamics in intelligent teaching analytics-supported collaborative reflection (Ucan & Webb, 2015). Examining shared regulation at different stages of intelligent teaching analytics implementation will provide a valuable insight into the dynamics of collaborative reflection. Therefore, our third research question was:

RQ3: How do pre-service teachers regulate their cognition, motivation, and emotions during intelligent teaching analytics-supported collaborative reflection?

To support the investigation of the research questions, the present study employed a design-based research (DBR) approach. DBR is a methodology designed by and for researchers that seeks to enhance the impact, transfer, and translation of education research into improved practice in teaching and learning (Anderson & Shattuck, 2012). This study is conducted in a real educational setting, specifically in collaborative reflection sessions among pre-service teachers, to explore effective methods and processes for applying intelligent teaching analytics. The researcher and pre-service teachers collaboratively design, implement and evaluate the effectiveness of different application methods and processes, refining them through multiple iterations to identify the optimal approach. DBR aligns well with the current study, as it emphasizes real educational context, collaboration between researchers and practitioners, and iterative testing—features widely utilized in prior research on the integration of technology in education (Chen & Oakley, 2020). A review of DBR in education revealed that 68% of interventions involve online and mobile technologies (Anderson & Shattuck, 2012), demonstrating the widespread use of DBR to apply technological tools in educational contexts.

Method

Context and participants

To address the research questions, this study collaborated with the Science Education Volunteer Service Association at a university in China. The association primarily consists of undergraduate students majoring in science education and educational technology, who are also pre-service teachers. Together, we designed an educational program to provide science lessons to schools with limited science teaching resources. The program aimed to enhance the teaching skills of pre-service teachers while also supplementing science education in these schools.

A total of 20 members of the association voluntarily participated in the study. Since the teaching plan required a minimum of seven pre-service teachers, the participation of 20 teachers was sufficient for successful implementation. Prior DBR studies in similar contexts have used comparable sample sizes (e.g., Wang et al., 2014a, 2014b; Zalavra et al., 2023). The participants (15 females, 5 males) were aged 17 to 21 and ranged from freshman to seniors. Three had prior classroom teaching experience, while the remaining 17 did not. Given the specific context and participants' motivations, their backgrounds should be considered when interpreting the research process and findings.

The volunteer teaching program consisted of seven sessions, with seven pre-service teachers serving as lead instructors. The remaining 13 participants in this project did not conduct teaching but primarily served as observers. They familiarized themselves with the teaching process by watching live teaching sessions or recorded videos and participated in collaborative reflection. One participant acted as both the observer and facilitator of the reflection meetings, while the other 12 served as observers. The facilitator and two observers attended every collaborative reflection session. The remaining 10 observers attended at least one session but were unable to attend all sessions due to university classes and personal commitments.

Material

This study employed the Yi-Class intelligent teaching analytics platform to analyze pre-service teachers' classroom practices, providing insights for teaching evaluation and fostering reflection. The platform provided various functions, including teaching mode analysis, teacher-student speech ratios, teacher attention analysis, and spatial positioning. It processed teaching videos using techniques such as computer vision and speech and language processing, generating data-driven results presented through charts and text explanations. The analysis results were exported as a PDF file, which we refer to as the "teaching analytics report". The platform was developed by our research team. The homepage of the Yi-class platform is shown in Fig. 1.

Procedure

Pre-service teachers attended a meeting where the first author introduced the intelligent teaching analytics platform. The meeting explained the functions of the platform, data collection methods, and analysis process. A week later, teaching sessions started taking place, followed by reflection meetings facilitated by the intelligent teaching analytics platform. The key steps of this reflective practice are illustrated in Fig. 2. During the weekly classroom teaching sessions, the class was recorded by cameras, and the videos were uploaded to the intelligent teaching analytics platform, which generated teaching analytics reports. Before the classroom teaching in the following week, all participants attended an in-person collaborative reflection meeting that lasted 1.5 to 2 hours.

The DBR approach was utilized to explore the application of intelligent teaching analytics in collaborative reflection. DBR typically involves identifying problems,



Fig.1 Homepage of the intelligent teaching analytics platform

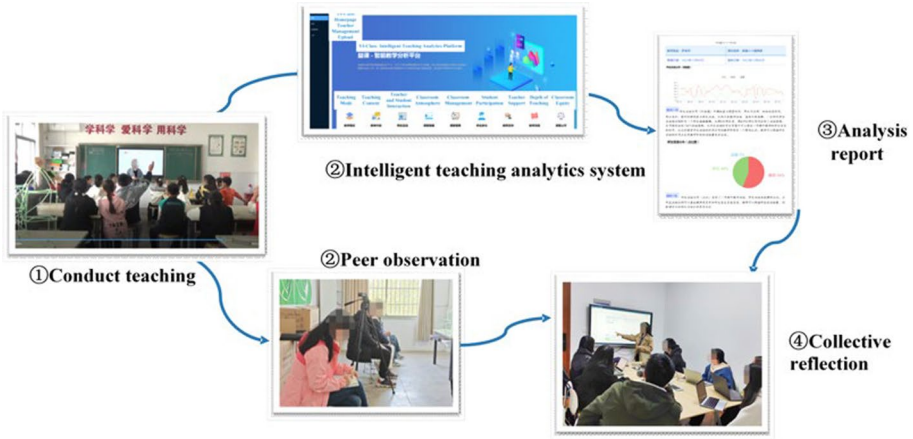


Fig.2 Teaching practice and reflection process

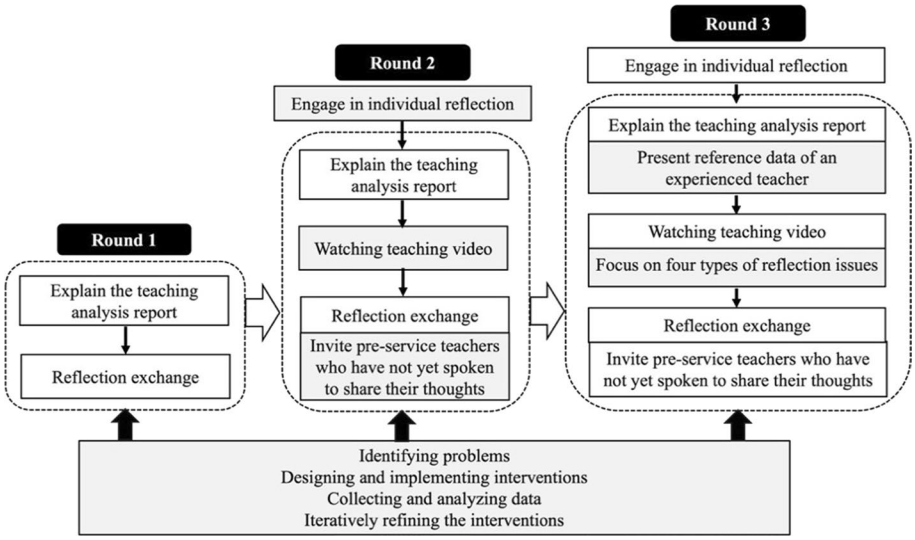


Fig.3 The procedure of three rounds of collaborative reflection

designing and implementing interventions, collecting and analyzing data, iteratively refining the interventions, and contributing to theory development (Onwuegbuzie et al., 2023). To identify problems, researchers conducted informal interviews with pre-service teachers after each collaborative reflection session, then discussed to develop the objectives and approach for applying teaching analytics in future sessions. By recording and analyzing data from the collaborative reflection sessions, an approach was provided for optimizing the next iteration. The procedure of three rounds application intervention is depicted in Fig. 3. The objectives, approaches, outcomes, and challenges of each round are detailed in Appendix 1. By the conclusion of the third round, we had successfully resolved several key challenges related to the application of intelligent teaching analytics in reflective practice. Although some new challenges were identified, they present valuable avenues for future investigation and refinement.

Data collection and analysis

The current study employed a mixed-methods approach, combining qualitative analysis of interview data with quantitative analysis of coding frequencies from collaborative reflection meeting transcripts. To address Research Questions 1 and 2, we conducted interviews with pre-service teachers following the completion of reflective practice activities. The rationale of the interview questions and their alignment with the research questions are detailed in Appendix 2. To address Research Question 3, we coded and analyzed transcripts from collaborative reflection sessions. Descriptive statistical analysis was used to compare the number of different shared regulation behavior codes, and lag sequence analysis (LSA) was adopted to examine the correlations and behavioral patterns between different behaviors. All pre-service teachers and cooperating schools were informed and gave their consent for the recordings. The video and audio recordings were used solely for reflective practice sessions and the current study.

Interview

Ten pre-service teachers completed the interview. To protect privacy, the pre-service teachers are referred to as PT1, PT2, …, PT10 in the remainder of the paper. Seven of them served as lead instructors conducting classroom teaching within the project (PT4-PT10), while the other three (PT1-3) were observers who participated in all collaborative reflection sessions. Among the observers, one also facilitated the reflection meetings (PT1), one had prior teaching experience (PT2), and one had no prior teaching experience (PT3). Detailed information about the interviewees is provided in Table 1.

The interview questions were collaboratively designed by the first author and a co-author, who is a professor in educational technology. To address the first research question, we asked two interview questions: one about perceptions of intelligent teaching analytics, and the other about suggestions for its improvement. To address the second research question, we asked two interview questions: one focused on the cognitive gains of pre-service teachers during intelligent teaching analytics supported collaborative reflection, and the other on their emotional experiences. In each reflection session, only one lead instructor, while the other six lead instructors attended as observers. Since most teachers served as both lead instructors and observers, we included two sub-questions for each role: one addressing their experience as lead instructors and the other as observers.

The interviews with pre-service teachers were conducted via online voice calls using a semi-structured format. Interviews followed the structured questions, with additional follow-up questions based on respondents' answers. Afterwards, the audio recordings were transcribed. The interview data was coded by two authors of this study using thematic analysis. Each coder independently completed the coding of responses from five interviewees. The coders first carefully read the interviewees' responses and conducted initial coding. Subsequently, they grouped similar codes to identify potential themes, which were then named and defined. Finally, the coders reviewed and discussed themes, reaching a consensus that they comprehensively and reliably captured the interviewees' responses.

Table 1 Basic information of participants

Participant ID	Gender	Classroom teaching experience before the project	Classroom teaching in the project	Role in collaborative reflection
PT1	Female	Yes	No	Observer and facilitator. Analyze the teaching process; Ensure focused discussion and active participation.
PT2	Male	Yes	No	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT3	Female	No	No	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT4	Male	Yes	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT5	Female	No	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT6	Female	No	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT7	Female	No	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT8	Male	No	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT9	Female	No	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.
PT10	Female	No	Yes	Observer. Analyze the teaching process, identify strengths and areas for improvement, and provide suggestions for teaching enhancement.

Collaborative reflection

Collaborative reflections were recorded using mobile phones and then the collaborative reflection recordings were transcribed into text using the speech-to-text tool "Xunjie Recording to Text Assistant". These transcripts included time stamps, speaker identities, and the content of their speech, totaling 3,692 entries. Shared regulation was the focus of to analysis in the collaborative reflection process, as it captures the core dynamics of collaborative reflection, illustrating how individuals collectively manage, regulate, and improve their learning and teaching practices across different stages of intelligent teaching analytics application. The coding framework for shared regulation strategies developed by Ucan and Webb (2015) was adopted, as it includes metacognitive, emotional and motivational categories, which can capture pre-service teachers' shared regulation behaviors in collaborative reflection activities.

Two authors of the paper jointly completed the coding process. Initially, the two coders familiar with the coding framework collaboratively coded 100 reflection texts. Subsequently, the coders independently coded 300 reflection texts in a back-to-back manner. The results indicated a coding consistency (Cohen's kappa) of 0.359, reflecting a low level of agreement. The coders then discussed and resolved discrepancies in their coding, clarifying their understanding of the coding standards and reaching a consensus. In the second round, the two coders again independently coded another 300 reflection texts back-to-back. This time, the Cohen's kappa value improved to 0.665, indicating moderate agreement. The coders subsequently revisited and clarified the remaining discrepancies, further refining their understanding of the coding standards. During the third round of coding, the two coders continued coding 300 reflection texts independently. The Cohen's kappa value in this round reached 0.76, indicating a higher level of agreement. At this stage, the coders were able to code more independently, while still engaging in discussions and negotiations over any discrepancies to ensure consistency in coding. The two coders independently coded the remaining 2,692 reflection texts, each coder independently tasked on 1,346 texts. To ensure reliability, they marked the uncertain codes in red during the coding process to facilitate discussion and achieve a consistent understanding through collaborative coding. A total of 3,663 valid codes were obtained.

LSA was applied to identify behavior patterns within the codes for shared regulation. LSA examines the probability of one behavior following another and its statistical significance, thereby uncovering underlying behavioral pattern (An & Zhang, 2024). GSEQ software was used to generate a frequency transition table, and Visio was used to create behavior transition diagrams based on the data.

Results

RQ1: What are pre-service teachers' perceptions and suggestions for intelligent teaching analytics?

Perceptions of intelligent teaching analytics

Pre-service teachers found intelligent teaching analytics reports echoed their teaching experience (PT4, PT5; e.g., PT4: "*I actually realized that during one part of my teaching, it was entirely me doing the lecturing. The report also pointed out the issue, noting that my sentences were too long and lacked clear pauses.*"). Intelligent teaching analytics reports also revealed insights they could not uncover independently (PT1, PT2, PT5, PT7, PT9, PT10; e.g., PT9: "*The analysis report helped me to discover unnoticed issues. I remember one particularly impactful insight: it pointed out that my class lacked interaction through analyzing the speaking frequency of both teachers and students. Most of the time, as a teacher, I was the one outputting information to the students without utilizing the class time for the students to process the knowledge they had learned.*"). Additionally, reports visually depict classroom performance, facilitating a thorough grasp of teaching effectiveness (PT3, PT7; e.g., PT7: "*My most direct impression is that the classroom performance of both teachers and students is presented using straightforward data and charts.*"). Two pre-service teachers stated that the evaluation metrics of intelligent teaching analytics compel them to make certain teaching data look better (PT4, PT10; e.g., PT10: "*From the analysis report, I can see the gap between myself and others, which*

motivates me to continuously improve the areas where my performance falls short."). In contrast, one pre-service teacher expressed difficulty understanding the analysis reports (e.g., PT8: *"I don' t understand why there is a need to emphasize "verbal exchanges between teachers and students."*).

Pre-service teachers generally found intelligent teaching analytics reports valuable, as the reports reflected their teaching experiences and uncovered issues that had previously gone unnoticed. The visual data helped them understand teaching effectiveness. Some teachers were motivated to improve, while others struggled to understand certain aspects of the reports, such as the emphasis on verbal exchanges. Overall, the analytics were perceived as insightful but occasionally difficult to interpret.

Suggestions for improving intelligent teaching analytics

Regarding the suggestions, one theme relates to the frame of reference. Some pre-service teachers indicated that the teaching content and the teaching approach demonstrated by the example class was different from their own, and as a result the analysis report of the example class had limited reference value for them. They suggest providing example classes that align with the teaching content they are currently covering (PT1, PT5, PT6, PT10; e.g., PT10: *"I think that example class that cover the same subject matter, teaching content, and teaching approach are more valuable as references."*), and include a variety of teaching approach (e.g., PT1: "*The teaching approach used in the example class are quite different and may not be appropriate for us to follow directly. Instead, I believe it would be more effective to select example classes that are widely recognized and accepted."*). **The second theme is relevant to expanding functions of intelligent teaching analytics**. This includes breaking down a lesson into its instructional phases and identifying issues within each phase (e.g., PT7: "*I think that if the analysis reports could break down our course into its instructional phases, it would more intuitively and accurately identify strengths and area for improvement at different stages."*), adding the analysis of students' attention, interest, and engagement in learning (e.g., PT9: "*Students' engagement and interest in learning are crucial for improving teaching. During teaching, it's sometimes difficult to monitor every student's status. Analyzing students' learning states would be very helpful for reflection."*), providing improvement strategies (PT5, PT8; e.g., PT5: "*Adding a strategy suggestions section would enable collaborative reflection to incorporate both peer feedback and recommendations from the intelligent system."*). **The third theme is about the visualization.** One participant indicated that analysis reports should be more concise and straightforward (e.g., PT9: "*The reports are a very detailed analysis. However, considering the time and effort required, I think these reports should include not only detailed analyses but also clear and concise summaries……Ideally, these suggestions should be listed in bullet points*."). **The fourth theme concerns with interactivity.** Some pre-service teachers suggested presenting the analysis reports as a shared document to enhance its interactivity in both presentation and usage (e.g., PT3: "*We could add interactive features to the reports, such as checking off items we agree with. This would make the reports more interactive and engaging, ensuring that everyone is more immersed while reading it. By encouraging deep reading and thorough understanding of the reports' content, we can better extract the valuable insights and discard the less useful parts.").*

RQ2: What are pre-service teachers' cognitive and emotional experiences of intelligent teaching analytics-supported collaborative reflection?

Cognitive experience during collaborative reflection

Pre-service teachers in different roles both report pedagogical knowledge gains and both focus on areas such as classroom management (PT4, PT5, PT7, PT8, PT9; e.g., PT8: "*I learned effective classroom management strategies, such as using commands, group work, and musical cues.*"; PT1, PT3, PT4, PT5, PT6, PT7; e.g., PT7: "*I found that a teacher's permissiveness could lead to discipline issues. Therefore, I strictly explained the classroom rules and order before the lesson, which made the lab session go smoothly.*"), organizational techniques (PT3, PT10; e.g., PT10: *"As an observer, I learned some teaching skills. For example, one member [teacher] asked students to place their lab materials in the center of the table after the experiment, which helped prevent distractions."*), student understanding(e.g., PT10: "I *gained a deeper understanding of students, especially the behavioral traits of them.*"), lesson design (e.g., PT7: "*I created a chart for students to fill out their experiment results, which was later displayed on the blackboard. This method was well-received and highlighted the feasibility of this approach, significantly influencing my future lesson planning.*") and use of the blackboard (PT6, PT7; e.g., PT6: "*During the reflection on the first lesson, we noticed that the teacher had omitted the use of the board, which resulted in a less comprehensive presentation of the lesson's key concepts.*").

However, there were also some issues. Several participants were also concerned about their ability to offer valuable suggestions and uncertain about the effectiveness of their feedback (PT1, PT4, PT9; e.g., PT1: "*At first, I didn't know how to observe the class and worried that my suggestions might not be very helpful.*"). While one teacher mentioned that she gradually learned to analyze teaching processes more systematically and identify potential issues (e.g., PT1: "*As I watched more lessons, I started by understanding the teaching design, and then observing the classroom instruction with specific questions and objectives in mind.*").

Pre-service teachers in different roles reported gains in pedagogical knowledge, particularly in classroom management, lesson design, organizational techniques, blackboard use, and understanding students. However, some pre-service teachers faced challenges in providing effective feedback and systematic observation. Over time, they developed stronger analytical skills to evaluate teaching processes.

Emotional experience during collaborative reflection

Most pre-service teachers (N = 7) experienced emotional changes over the course of collaborative reflection sessions. This was particularly true for those who served as lead instructors. Several participants initially felt nervous and embarrassed (PT9, PT10; e.g., PT9: "*I felt slightly awkward because I believed my teaching process wasn't very good.*" PT10: "*As an instructor, I felt nervous and feared that I might give a poor lesson and face criticisms.*"). For those in the observer role, there was initially a general sense of reservation (e.g., PT4: "*At first, we were quite unfamiliar with each other. Everyone tended to emphasize positive aspects, which is good, but there were fewer mentions of weaknesses.*"). As the reflection activities progress, the familiarity among participants increases, and pre-service teachers felt that the atmosphere of reflection became more equitable and

lively (PT1, PT3, PT5, PT6, PT9, PT10; e.g., PT1: "*I think engaging in teaching practice and reflection together is a gradual process of improvement. After all, it's a journey from being unfamiliar to becoming quite familiar with each other, and we are now doing things together. So, I feel that the atmosphere is gradually getting better and more balanced.*"). Overall, most participants (N = 9) had a positive evaluation of the reflection atmosphere, considering it friendly and describing the relationships among members as harmonious, enthusiastic, and mutually supportive (PT1, PT2, PT4, PT5, PT6, PT7, PT8, PT9, PT10; e.g., PT10: "*It is a harmonious, mutually progressive relationship. This relationship becomes even stronger after collaborative reflection.*").

RQ3: How do pre-service teachers regulate their cognition, motivation, and emotions during intelligent teaching analytics-supported collaborative reflection?

Table 2 present the descriptive statistics. It can be seen that the planning (MP) behavior significantly increased to 14.1% in R3, reflecting the importance of re-planning task objectives R3. Monitoring (MM) behavior has always had the highest proportion, but it has decreased in R3 (from 81.5% in R1 to 70.7% in R3), indicating that as the work on the task progressed, pre-service teachers' demand for monitoring weakened. The proportion of evaluation (ME) behaviors gradually increased from 6.9% in R1 to 11.6% in R3, indicating that members paid more attention to reflection and summarization in R3. Emotion and motivation regulation monitoring (EMM) and control (EMC) behaviors were most prevalent in R2. The lower levels observed in R1 may be due to pre-service teachers

Table 2 Descriptive statistics of shared regulation codes

Category	Codes	R1	R2	R3
Metacognitive regulation Planning (MP)	GS	5	4	23
	CC	10	5	9
	PT	26	59	204
	AN	4	9	3
	Total	45	77	239
	Percentage	8.9%	5.3%	14.1%
Metacognitive regulation monitoring (MM)	AU	277	792	736
	MAP	15	20	73
	QfU	82	272	301
	SC	39	156	86
	Total	413	1240	1196
	Percentage	81.5%	84.7%	70.7%
Metacognitive regulation evaluation (ME)	RLP	35	67	196
	Total	35	67	196
	Percentage	6.9%	4.6%	11.6%
Emotional and motivational regulation monitoring (EMM)	AE	4	17	17
	AM	0	14	14
	Total	4	31	31
	Percentage	0.8%	2.1%	1.8%
Emotional and motivational regulation control (EMC)	EH	2	10	12
	PC	8	39	28
	Total	10	49	30
	Percentage	2.0%	3.3%	1.8%

being in the adaptation phase, while the decline in R3 could be attributed to the task approaching its conclusion.

The results of the LSA for the first round are shown in Fig. 4. In the figure, the nodes represent the coded behaviors, and the lines indicate significant connections between behaviors. The numbers on the lines represent the adjusted residual Z-scores, and the thickness of the lines varies according to the adjusted residual Z-scores. The larger the adjusted residual Z-score, the thicker the line, indicating a higher significance between the behavior. In the first round, nine significant behavior sequences were identified. In terms of behavior transition frequency, the highest frequency was observed in the self-transition behavior of goal setting (GS → GS, 6.78), followed by the allocation of roles and negotiation of task responsibilities (AN → AN, 5.04). Although the total number of GS and AN codes was not the highest, their elevated transition frequency in the behavior sequence highlights their crucial role in connecting key behavior within the group interaction. Additionally, reviewing or evaluating the teaching and reflection process also exhibited relatively high transition rates (RLP → RLP, 4.19). The sequence EH → SC (3.43) suggests that encouragement is given before pointing out others' mistakes and misunderstandings, reflecting a control over motivation. Similarly, the sequence PC → GS (3.24) indicates that praise or compliments are used before setting goals, demonstrating a pattern of using motivation control to advance cognitive processes.

The results of the LSA for the second round are shown in Fig. 5. In the second round, 13 significant behavior sequences were formed, with the highest frequency observed in monitoring the reflection process to judge whether the goal was completed (MAP → MAP, 9.16), followed by reviewing or evaluating the group's reflection process (RLP → RLP, 8.93), and lastly, the planning category, which involved assigning roles and negotiating task responsibilities (AN → AN, 8.32). Within the planning category, PT played a central role, not only exhibiting self transition (PT → PT, 3.8) but also being closely linked to goal setting (PT → GS, 4.68), clarifying conditions (PT → CC, 4.1), and negotiating tasks and roles (PT → AN, 2.78). Additionally, all five categories were present in the second round, with emotional monitoring behaviors particularly standing out, such as the self transition of awareness of emotion (AE → AE, 4.1) and the praise and encouragement that followed emotion awareness (AE → PC, 2.34). This indicates that emotional regulation not only focused on recognizing and regulate emotions but also on fostering motivation. However,

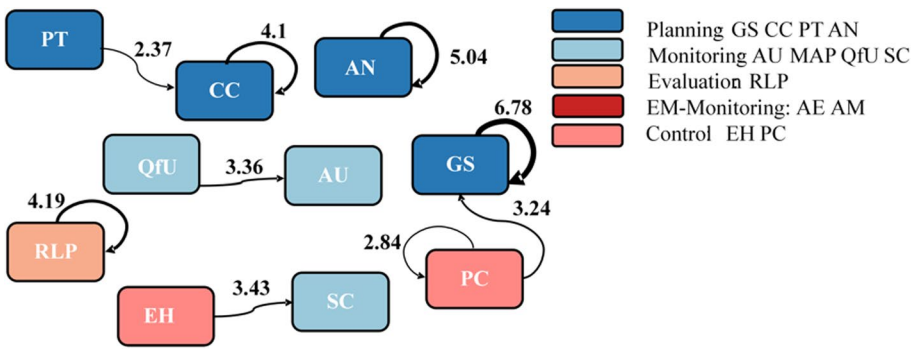


Fig.4 Analysis of shared regulation patterns in the first round of reflection

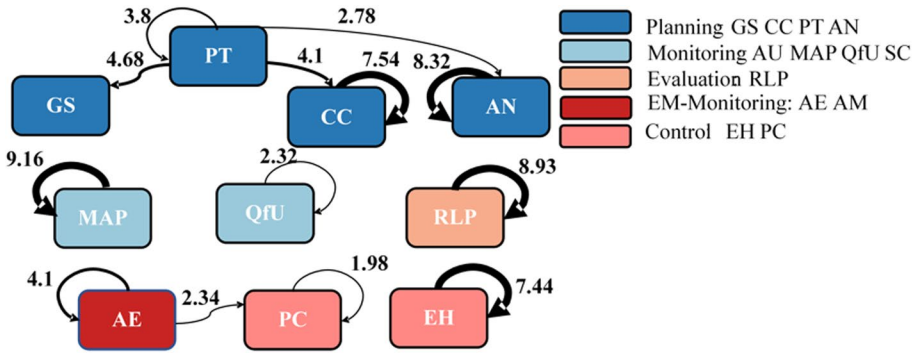


Fig.5 Analysis of shared regulation patterns in the second round of reflection

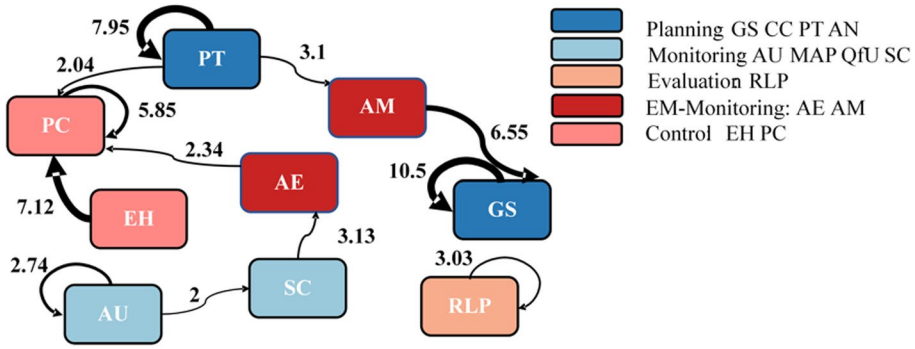


Fig.6 Analysis of shared regulation patterns in the third round of reflection

there were nine behaviors that only exhibited self transition, suggesting that the regulation pattern in the shared regulation process remained relatively linear.

The results of the LSA for the third round are shown in Fig. 6. In the third round, 12 behavior sequences were significant. Similar to the first round, goals setting emerged as the most frequent behavior (GS → GS, 10.5), followed by asking or deciding how to execute tasks (PT → PT, 7.95). Emotion and motivation control become manifest with encouragement followed by praise and compliments (EH → PC, 7.12). As in the second round, all five behavior categories appeared; however, what stood out in the third round was the increase in emotional and motivational behaviors, which formed significant behavior sequences with cognitive codes. For example, PT → AM → GS (3.1,6.55) and PT → PC (2.04) were observed. This shift indicates a growing awareness of interpersonal dynamics, highlighting that pre-service teachers began incorporating motivation into task planning process. Moreover, the third round showed only four behavior with self transition. This demonstrates that the connections between different behaviors became more intricate, signaling a more developed and enriched shared regulation process during reflection.

Discussion and implications

Discussion

Regarding perception, pre-service teachers found intelligent teaching analytics reports valuable for reflecting on their teaching and identifying areas for improvement. This indicates

acceptance of intelligent teaching analytics among pre-service teachers. The result can be attributed to several factors. First, younger age people generally show a higher propensity to adopt new technologies, as supported by existing research (Park et al., 2019; Yang et al., 2019). Second, pre-service teachers' acceptance of technology may also be positively influenced by their peers. The Unified Theory of Acceptance and Use of Technology (UTAUT) suggests that social influence is one of the four constructs that directly determine behavioral intention (Williams et al., 2015). Third, pre-service teachers are motivated to improve their teaching, which makes them more likely to accept tools that align with their goal of enhancing teaching performance. This reason is supported by previous studies, which indicate that when the perceived value of a tool aligns with users' needs and motivations, it positively influences technology adoption (Khechine et al., 2020).

However, some pre-service teachers struggled to interpret certain aspects of the teaching analytics reports. Similar challenges were also observed in learning analytics (Tsai & Gasevic, 2017). This result may be attributed to two main factors. First, the data-driven analysis often condenses rich, process-oriented information, making it harder to interpret. Second, it is linked to pre-service teacher's data literacy (Shreiner & Dykes, 2021). Pre-service teachers need to possess data literacy to understand and interpret the data and visualization generated by intelligent teaching analytics. As Ndukwe and Daniel emphasized, data literacy is essential to enable educators to effectively engage with data and analytics to enhance their pedagogical decisions.

Observers, like lead instructors, reported gains in pedagogical knowledge, with both roles highlighting areas such as classroom management, student understanding and blackboard use. This supports the effectiveness of observational learning, aligning with previous research and practices that emphasize its importance in teacher training and development (Frei-Landau et al., 2022). Regarding emotions, both observers and lead instructors reported a more open and harmonious reflective atmosphere over time, with increased familiarity and connection among members—an aspect often identified in previous research as a sign of community evolution (Wenger et al., 2002). This reflects the organizers' emphasis on developmental reflection as well as participants' emotional and motivational regulation, as suggested by the DBR implementation and shared regulation analysis. Unsurprisingly, emotional experiences varied by role: lead instructors tended to worry about criticism and felt embarrassed or ashamed when teaching problems were identified, while observers were primarily concerned with providing insufficient feedback. Role theory (Biddle, 1979) offers a valuable perspective to understand these differences, linking them to the unique responsibilities associated with each role. Similarly, previous studies found that teachers perceive varying levels of support in technology-integrated educational settings based on their roles (Lin et al., 2024).

The shared regulation process exhibited distinct characteristics across the three rounds, shaped by both the optimization of the application approach and the effects of time. LSA results revealed that in the first round, pre-service teachers predominantly focused on goal setting (GS → GS) and task allocation (AN → AN), indicating an early emphasis on structured planning and role distribution. This aligns with prior research suggesting that initial phases of collaborative learning often prioritize planning and role assignment (Liu et al., 2018). Although emotional regulation behaviors were present in the first round (EMM = 4, EMC = 8), they did not form significant behavioral sequences with other behavior or

themselves. This absence of meaningful connections likely reflects pre-service teachers' greater focus on task execution over emotional interaction at early stage. In the second and third rounds, there was a notable increase in emotional and motivational regulation sequences, particularly encouragement and praise (AE → PC). This supports cognitive reappraisal that reflection allowing individuals to reinterpret emotional experiences positively, which is crucial for effective emotional regulation (Arditte & Joormann, 2011). Despite this improvement, some behaviors continued to exhibit self-transition, indicating that shared regulation had not yet fully matured. This may be due to the study's emphasis on optimizing the application process rather than directly fostering shared regulation development.

Implications

This research contributes to the existing literature on intelligent teaching analytics and pre-service teacher development by integrating concepts of intelligent teaching analytics, data literacy, and theories of shared regulation, while also uncovering the cognitive and emotional experiences these processes can foster. Based on the findings of this study, we offer six principles for the application of intelligent teaching analytics. First, engage in individual reflection before moving to collaborative reflection. Second, focus on specific aspects of teaching, such as classroom management or student interactions, as broad topics would hinder deeper analysis. Third, provide teaching videos for facilitating the understanding of analytics results. Fourth, present reference data from experienced teachers in similar courses and contexts to serve as a frame of reference. Fifth, develop teaching improvement strategies by collaborating with educational experts or by leveraging generative large language models, tailored to address the instructional issues identified through reflection. Sixth, build a safe, open and encouraging environment for authentic and critical reflection.

When placing the current field study within a broader context, it serves as a high-fidelity prototype that advances intelligent teaching analytics to foster collaborative reflection, offering valuable insights into larger-scale applications. This study involves crucial stakeholders such as pre-service teachers, technology developers, experienced teachers, and researchers, all of which are interconnected in promoting intelligent teaching analytics-supported collaborative reflection. Thus, we proposed future directions for each of these roles.

(1) For teacher education institutions, strengthen training in data literacy (Shreiner & Dykes, 2021; Tsai & Gašević,2017) among pre-service teachers to ensure they can understand and interpret the findings of intelligent teaching analytics, fostering evidence-based reflection and making data-driven instructional decisions.

(2) Regarding technology developers, further optimization and expansion of the functions of intelligent teaching analytics are needed. For example, the teaching process could be decomposed into different instructional events, with tailored intelligent analysis for a more granular examination of the teaching process. Additionally, integrating analysis of the student learning process, adding a section dedicated to improvement suggestions, and enhancing the interactivity of the analysis report.

(3) For education researchers, designing and implementing intervention strategies based on the expanded Technology Acceptance Model (Wang et al., 2025) and feedback intervention theory (Kluger & DeNisi, 1996) to enhance the acceptance and effectiveness of intelligent teaching analytics feedback. Additionally, for specific learning contexts such as collaborative reflection, scaffolds should be designed with reference to socially shared regulation theory (Hadwin et al., 2017) to improve the effectiveness of collaborative reflection.

(4) A collaborative effort among educational researchers, expert teachers, and technology specialists should be made to develop a teaching strategy repository. This platform would offer pre-service teachers access to tailored instructional strategies, helping them address common classroom challenges. By incorporating expert guidance, practical examples, and peer feedback, it would support continuous professional development and enhance teaching practices.

Conclusion

To bridge the research gap in applying intelligent teaching analytics to pre-service teachers' collaborative reflection, this study used DBR to iteratively improve the application process, describe the evolutionary process of collaborative reflection, and reveal its impact on pre-service teachers. This study demonstrates the potential of intelligent teaching analytics to enhance teaching reflection, with the following contributions: (1) Proposes principles for the application of intelligent teaching analytics to facilitate collaborative reflection; (2) Explores the cognitive and emotional impacts on pre-service teachers of applying intelligent teaching analytics in collaborative reflection; (3) By situating this study as a high-fidelity prototype within a broader application context, it offers recommendations for teacher training institutions, technology developers, teaching experts and education researchers.

Despite the contributions of this research, a limitation should be noted: the 7-week duration dictated by the project timeline may limited the opportunity to fully explore its potential long-term impacts. A longer-term study would provide pre-service teachers with more opportunities for teaching practice and reflection, while also allowing for continuous adjustments to the application of intelligent teaching analytics based on teachers' needs and different stages of their development. Therefore, future research could build on this study by extending the study period to explore further refinements in the application of intelligent teaching analytics. Additionally, integrating large language models, such as ChatGPT, with teaching analytics to assess their influence on teachers' collaborative reflection could offer valuable insights. This integration may enhance the feedback process and provide more tailored guidance, ultimately supporting teachers in improving their instructional practices.

Appendixces

Appendix 1: Three round iteration overviews

Time	Objectives	Approaches	Outcomes	Challenges
First Round (week 1)	Introduce the intelligent teaching analytics report and encourage pre-service teachers to reflect on their teaching in conjunction with the report. Create a positive and safe atmosphere for reflection.	Present and introduce the intelligent teaching analytics report. Ask pre-service teachers about their understanding of the analysis report and how they reflect upon seeing it. Encourage teachers to adopt a growth mindset by affirming their teaching practices and framing challenges as opportunities for development.	Lead instructors, facilitators, and senior observers actively participated in the collaborative reflection process.	Most other participants demonstrated lower levels of engagement. The majority of contributions came from the lead teacher, organizer, and experienced individuals, with other pre-service teachers speaking less frequently. There was also a challenge in aligning the chronological data in the analysis report with the corresponding teaching moments. Additionally, the absence of video recordings to support the intelligent teaching analytics reports made it difficult to verify and fully understand the findings.
Second Round (Weeks 2–4)	Foster observer engagement: Implement measures to ensure active participation from all observers. Bolster analysis credibility: Strengthen the analysis report by incorporating diverse perspectives and corroborative evidence.	Require reflective writing: Require all participants to submit written reflections on their teaching experiences. Proactively solicit reflections: Encourage organizers to actively seek input from those who have not yet contributed. Utilize teaching videos: Introduce teaching videos to support and validate the findings in the analysis report during discussions.	The number of participants contributing during collaborative reflection increased.	Reflections on the teaching videos lacked specific focus, resulting in low efficiency. Pre-service teachers' reflections were fragmented, and discussions on the content were not sufficiently in-depth, often failing to reach solutions. Pre-service teachers struggled to assess the quality of teaching based on the analysis report, as they were uncertain which data indicated areas needing improvement.
Third Round (Weeks 5–7)	Focus reflections on key issues to ensure they are purposeful. Provide reference data on teaching performance.	Categorize common teaching issues from past practices into four areas: classroom management, classroom interaction, experimental organization, and teaching style. Use these categories to identify problems in teaching videos and discuss corresponding solutions. Analyze a video of an experienced science teacher, using the average and extreme data values from multiple classes as reference frame.	This led to more focused collaborative reflection processes and deeper discussions. Pre-service teachers understood the gap between themselves and the reference teacher.	Some pre-service teachers were not entirely convinced by the target reference data. They expressed a desire for the analysis report to provide more strategies and suggestions for bridging these gaps.

Appendix 2: Rationale of the designed interview question and alignment of interview question with research questions

Research questions	Interview questions	Rationale of the designed interview question and alignment of interview question with research questions
RQ1: What are pre-service teachers' perceptions and suggestions for intelligent teaching analytics?	1.1 What are your perceptions of intelligent teaching analytics in the collaborative reflection you have experienced? 1.2 What suggestions do you have for improving intelligent teaching analytics to better support reflection?	[Rationale] Question 1.1 aims to understand pre-service teachers' perceptions of intelligent teaching analytics. Question 1.2 focuses on optimizing intelligent teaching analytics further. [Alignment] Question 1.1 addresses the "perception" aspect of the research question. Question 1.2 addresses the "suggestion" aspect of the research question.
RQ2: What are pre-service teachers' cognitive and emotional experiences during intelligent teaching analytics-supported collaborative reflection?	2.1 When you were the lead instructor, did you gain any cognitive benefits during collaborative reflection supported by intelligent teaching analytics? If so, please describe them in detail. 2.2 When you were an observer, did you gain any cognitive benefits during collaborative reflection supported by intelligent teaching analytics? If so, please describe them in detail. 2.3 When you were the lead instructor, what were your emotional experiences during collaborative reflection supported by intelligent teaching analytics? Please describe them in detail. 2.4 When you were an observer, what were your emotional experiences during collaborative reflection supported by intelligent teaching analytics? Please describe them in detail.	[Rationale] The cognitive outcomes explore whether collaborative reflection supported by intelligent teaching analytics benefits pre-service teachers' cognition. The emotional experience examines whether the organization of collaborative reflection in this context positively impacts pre-service teachers' emotions. Since participants assume different roles, their cognitive experiences are likely to vary. To address this, sub-questions were designed to distinguish between roles. Differences in roles may lead to distinct cognitive experiences; Similarly, the emotional experiences were also analyzed based on roles. Learners' cognitive and emotional experiences with technology naturally occur during its use. Intelligent teaching analytics are embedded within the context of collaborative reflection. As collaborative reflection incorporates intelligent teaching analytics, it is difficult to isolate whether the observed cognitive and emotional outcomes stem solely from intelligent teaching analytics or from the collaborative reflection supported by it. Therefore, our investigation focuses on the cognitive and emotional experiences within collaborative reflection supported by intelligent teaching analytics. [Alignment] Questions 2.1 and 2.2 address the "cognitive experience" in the research question, while Questions 2.3 and 2.4 focus on the "emotional experience".

Acknowledgements
We would like to thank all the participants involved in the study. We also appreciate Justin Edwards for his valuable feedback and language editing assistance.

Author contributions
Mengke Wang: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing—original draft and review, Visualization, Project administration; Zengzhao Chen: Supervision, Funding acquisition, Project administration; Ying Xu: Formal analysis, Investigation, Data curation, Project administration; Bhagya Maheshi: Methodology, Writing—review and editing; Dragan Gašević: Conceptualization, Supervision, Writing—review and editing, Funding acquisition.

Funding
Funding was provided by National Natural Science Foundation of China (62077022), China Scholarship Council (CSC202306770066), Independent Research Project of Central China Normal University (CCNU24ai013), Australian Research Council (DP210100060, DP240100069)

Availability of data and materials
The datasets and material used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
All participants agreed to take part in this study and provided written consent.

Competing interests
The authors declare that they have no competing interests.

Received: 8 October 2024 Accepted: 9 May 2025
Published online: 21 July 2025

References

Ahuja, K., Kim, D., Xhakaj, F., Varga, V., Xie, A., Zhang, S., Townsend, J. E., Harrison, C., Ogan, A., & Agarwal, Y. (2019). Edusense: Practical classroom sensing at scale. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 3*(3), 1–26.

An, S., & Zhang, S. (2024). Effects of ability grouping on students' collaborative problem solving patterns: Evidence from lag sequence analysis and epistemic network analysis. *Thinking Skills and Creativity, 51*, Article 101453.

Anderson, T., & Shattuck, J. (2012). Design-based research: A decade of progress in education research? *Educational Researcher, 41*(1), 16–25.

Arditte, K. A., & Joormann, J. (2011). Emotion regulation in depression: Reflection predicts recovery from a major depressive episode. *Cognitive Therapy and Research, 35*, 536–543.

Bandura, A. (1997). *Self-efficacy: The exercise of control*. Macmillan.

Barsade, S. G., Coutifaris, C. G., & Pillemer, J. (2018). Emotional contagion in organizational life. *Research in Organizational Behavior, 38*, 137–151.

Biddle, B. J. (1979). *Role theory: Expectations, identities, and behaviors*. Academic Press.

Bond, M. (2024). The International Journal of Educational Technology in Higher Education: Content and authorship analysis 2010–2024. *International Journal of Educational Technology in Higher Education, 21*(1), 60.

Burrell, J. (2016). How the machine 'thinks': Understanding opacity in machine learning algorithms. *Big Data & Society*. https://doi.org/10.1177/2053951715622512

Cai, H., Lu, L., Han, B., Wong, L. H., & Gu, X. (2024). Exploring pre-service teachers' reflection mediated by an AI-powered teacher dashboard in video-based professional learning: a pilot study. *Educational technology research and development*, 1–26.

Celepkolu, M., Wiggins, J. B., Galdo, A. C., & Boyer, K. E. (2021). Designing a visualization tool for children to reflect on their collaborative dialogue. *International Journal of Child-Computer Interaction, 27*, Article 100232.

Chen, K. Z., & Oakley, B. (2020). Redeveloping a global MOOC to be more locally relevant: Design-based research. *International Journal of Educational Technology in Higher Education, 17*, 1–22.

Clarà, M., Mauri, T., Colomina, R., & Onrubia, J. (2019). Supporting collaborative reflection in teacher education: A case study. *European Journal of Teacher Education, 42*(2), 175–191.

De France, K., & Hollenstein, T. (2022). The Development of Cognitive Reappraisal for Regulating Emotions. The Oxford Hand-book of Emotional Development, *Oxford Academic*, 110–126.

Echeverria, V., Yan, L., Zhao, L., Abel, S., Alfredo, R., Dix, S., Jaggard, H., Wotherspoon, R., Osborne, A., Buckingham Shum, S., & Martinez-Maldonado, R. (2024). Teamslides: a multimodal teamwork analytics dashboard for teacher-guided reflection in a physical learning space. *Proceedings of the 14th Learning Analytics and Knowledge Conference* (pp. 112–122).

Frei-Landau, R., Orland-Barak, L., & Muchnick-Rozonov, Y. (2022). What's in it for the observer? Mimetic aspects of learning through observation in simulation-based learning in teacher education. *Teaching and Teacher Education, 113*, Article 103682.

Gašević, D., Dawson, S., & Siemens, G. (2015). Let's not forget: Learning analytics are about learning. *TechTrends, 59*, 64–71.

Hadwin, A., Järvelä, S., & Miller, M. (2017). Self-regulation, co-regulation, and shared regulation in collaborative learning environments. In *Handbook of Self-regulation of Learning and Performance* (pp. 83–106). Routledge.
Hao, Y. (2016). The development of pre-service teachers' knowledge: A contemplative approach. *Computers in Human Behavior, 60*, 155–164.
Huang, C., Wu, X., Wang, X., He, T., Jiang, F., & Yu, J. (2021). Exploring the relationships between achievement goals, community identification and online collaborative reflection. *Educational Technology & Society, 24*(3), 210–223.
Huang, X., & Lajoie, S. P. (2023). Social emotional interaction in collaborative learning: Why it matters and how can we measure it? *Social Sciences & Humanities Open, 7*(1), Article 100447.
Jacobs, J., Scornavacco, K., Harty, C., Suresh, A., Lai, V., & Sumner, T. (2022). Promoting rich discussions in mathematics classrooms: Using personalized, automated feedback to support reflection and instructional change. *Teaching and Teacher Education, 112*, Article 103631.
Järvelä, S., Kirschner, P. A., Hadwin, A., Järvenoja, H., Malmberg, J., Miller, M., & Laru, J. (2016). Socially shared regulation of learning in CSCL: Understanding and prompting individual-and group-level shared regulatory activities. *International Journal of Computer-Supported Collaborative Learning, 11*, 263–280.
Jay, J. K., & Johnson, K. L. (2002). Capturing complexity: A typology of reflective practice for teacher education. *Teaching and Teacher Education, 18*(1), 73–85.
Johnson, M. (2016). Feedback effectiveness in professional learning contexts. *Review of Education, 4*(2), 195–229.
Kelly, P. (2006). What is teacher learning? A social-cultural perspective. *Oxford Review of Education, 32*(40), 505–519.
Khechine, H., Raymond, B., & Augier, M. (2020). The adoption of a social learning system: Intrinsic value in the utaut model. *British Journal of Educational Technology, 51*(6), 2306–2325.
Khosravi, H., Shum, S. B., Chen, G., Conati, C., Tsai, Y.-S., Kay, J., Knight, S., Martinez-Maldonado, R., Sadiq, S., & Gašević, D. (2022). Explainable artificial intelligence in education. *Computers and Education: Artificial Intelligence, 3*, Article 100074.
Killeavy, M., & Moloney, A. (2010). Reflection in a Social Space: Can blogging support reflective practice for beginning teachers? *Teaching and Teacher Education, 26*, 1070–1076. https://doi.org/10.1016/j.tate.2009.11.002
King, F. (2014). Evaluating the impact of teacher professional development: An evidence-based framework. *Professional Development in Education, 40*(1), 89–111.
Kizilcec, R. F. (2016). How much information? Effects of transparency on trust in an algorithmic interface. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 2390–2395.
Kluger, A. N., & DeNisi, A. (1996). The effects of feedback interventions on performance: A historical review, a meta-analysis, and a preliminary feedback intervention theory. *Psychological Bulletin, 119*(2), 254.
Lee, U., Jeong, Y., Koh, J., Byun, G., Lee, Y., Lee, H., Eun, S., Moon, J., Lim, C., & Kim, H. (2024). I see you: teacher analytics with GPT-4 vision-powered observational assessment. *Smart Learning Environments, 11*(1), 48.
Lin, X. F., Chiu, T. K., Luo, S., Wong, S. Y., Hwang, H., Hwang, S., Li, W., Liang, Z. M., Peng, S., & Lin, W. (2024). Teacher learning community for AR-integrated STEM education. *Teaching and Teacher Education, 141*, Article 104490.
Liu, Y. H., Kwon, K., & Johnson, L. P. (2018). Exploration of factors in the early collaboration phase affecting virtual groups' overall collaborative learning experiences. *Journal of Educational Computing Research*, 56(4), 485–512.
Martinez-Maldonado, R., Echeverria, V., Fernandez-Nieto, G., Yan, L., Zhao, L., Alfredo, R., Li, X., Dix, S., Jaggard, H., Wotherspoon, R., et al. (2023). Lessons learnt from a multimodal learning analytics deployment in-the-wild. *ACM Transactions on Computer-Human Interaction, 31*(1), 1–41.
Martinez-Maldonado, R., Echeverria, V., Mangaroska, K., Shibani, A., Fernandez-Nieto, G., Schulte, J., & Buckingham Shum, S. (2022). Moodoo the tracker: Spatial classroom analytics for characterising teachers' pedagogical approaches. *International Journal of Artificial Intelligence in Education, 32*, 1–27.
McConchie, J., Hite, B. J., Blackard, M. B., & Cheung, R. C. M. (2022). With a little help from my friends: Development and validation of the positive peer influence inventory. *Applied Developmental Science, 26*(1), 74–93.
Miller, M., & Hadwin, A. (2015). Scripting and awareness tools for regulating collaborative learning: Changing the landscape of support in CSCL. *Computers in Human Behavior, 52*, 573–588.
Ndukwe, I. G., & Daniel, B. K. (2020). Teaching analytics, value and tools for teacher data literacy: A systematic and tripartite approach. *International Journal of Educational Technology in Higher Education, 17*, 1–31.
Onwuegbuzie, A. J., Forzani, E., & Corrigan, J. A. (2023). A fully integrated systematic review of mixed methods design-based research. *Journal of Mixed Methods Studies, 8*, 24–51.
Park, C., Kim, D. G., Cho, S., & Han, H. J. (2019). Adoption of multimedia technology for learning and gender difference. *Computers in Human Behavior, 92*, 288–296.
Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review, 18*, 315–341.
Pfitzner-Eden, F. (2016). Why do I feel more confident? Bandura's sources predict preservice teachers' latent changes in teacher self-efficacy. *Frontiers in Psychology, 7*, 1486.
Poonja, H. A., Shirazi, M. A., Khan, M. J., & Javed, K. (2023). Engagement detection and enhancement for stem education through computer vision, augmented reality, and haptics. *Image and Vision Computing, 136*, Article 104720.
Pozdniakov, S., Martinez-Maldonado, R., Tsai, Y. S., Echeverria, V., Srivastava, N., & Gasevic, D. (2023, March). How do teachers use dashboards enhanced with data storytelling elements according to their data visualisation literacy skills?. In *LAK23: 13th International Learning Analytics and Knowledge Cnference* (pp. 89–99).
Prilop, C. N., Weber, K. E., & Kleinknecht, M. (2021). The role of expert feedback in the development of pre-service teachers' professional vision of classroom management in an online blended learning environment. *Teaching and Teacher Education, 99*, Article 103276.
Qazi, A. G., & Pachler, N. (2024). Conceptualising a data analytics framework to support targeted teacher professional development. *Professional Development in Education*, 1–24.
Ramakrishnan, A., Zylich, B., Ottmar, E., LoCasale-Crouch, J., & Whitehill, J. (2021). Toward automated classroom observation: Multimodal machine learning to estimate class positive climate and negative climate. *IEEE Transactions on Affective Computing, 14*(1), 664–679.

Ross, J. (2011). Traces of self: Online reflective practices and performances in higher education. *Teaching in Higher Education, 16*(1), 113–126.
Segal, A. (2024). Rethinking collective reflection in teacher professional development. *Journal of Teacher Education, 75*(2), 155–167.
Shreiner, T. L., & Dykes, B. M. (2021). Visualizing the teaching of data visualizations in social studies: A study of teachers' data literacy practices, beliefs, and knowledge. *Theory & Research in Social Education, 49*(2), 262–306.
Shuo, W., Zengzhao, C., Mengke, W., Yawen, S., & Shenghu, Z. (2022). Teacher attention measurement based on head pose estimation. In *2022 International Conference on Intelligent Education and Intelligent Research* (IEIR) (pp. 1–7). IEEE.
Sorensen, P. (2014). Collaboration, Dialogue and Expansive Learning: The Use of Paired and Multiple Placements in the School Practicum. *Teaching and Teacher Education, 44*, 128–137. https://doi.org/10.1016/j.tate.2014.08.010
Sümer, Ö., Goldberg, P., D'Mello, S., Gerjets, P., Trautwein, U., & Kasneci, E. (2021). Multimodal engagement analysis from facial videos in the classroom. *IEEE Transactions on Affective Computing, 14*(2), 1012–1027.
Tan, S. E., & Jung, I. (2024). Unveiling the dynamics and impact of emotional presence in collaborative learning. *International Journal of Educational Technology in Higher Education, 21*(1), 44.
Tillema, H., & Van der Westhuizen, G. J. (2006). Knowledge construction in collaborative enquiry among teachers. *Teachers and Teaching: Theory and Practice,* 12 (1): 51–67. https://doi.org/10.1080/13450600500365403
Treacy, M., & Leavy, A. (2023). Student voice and its role in creating cognitive dissonance: The neglected narrative in teacher professional development. *Professional Development in Education, 49*(3), 458–477.
Tsai, Y. S., & Gašević, D. (2017, March). Learning analytics in higher education---challenges and policies: a review of eight learning analytics policies. *Proceedings of the Seventh International Learning Analytics & Knowledge Conference* (pp. 233–242).
Tsang, K. K. (2011). Emotional labor of teaching. *Educational Research, 2*(8), 1312–1316.
Ucan, S., & Webb, M. (2015). Social regulation of learning during collaborative inquiry learning in science: How does it emerge and what are its functions? *International Journal of Science Education, 37*(15), 2503–2532.
Unal, E., & Cakir, H. (2021). The effect of technology-supported collaborative problem solving method on students' achievement and engagement. *Education and Information Technologies, 26*(4), 4127–4150.
UNESCO. (2024). AI framework for teacher competency. United Nations Educational, Scientific and Cultural Organization. Retrieved from https://www.unesco.org/en/articles/ai-competency-framework-teachers.
van Geel, M., Keuning, T., Visscher, A., & Fox, J.-P. (2017). Changes in educators' data literacy during a data-based decision making intervention. *Teaching and Teacher Education, 64*, 187–198.
van Leeuwen, A., Teasley, S. D., & Wise, A. (2022). Teacher and student facing analytics. In *Handbook of learning analytics* (pp. 130-140). Society for Learning Analytics Research.
Vygotsky, L. S., & Cole, M. (1978). *Mind in society: Development of higher psychological processes*. Harvard University Press.
Wang, S. K., Hsu, H. Y., Reeves, T. C., & Coster, D. C. (2014a). Professional development to enhance teachers' practices in using information and communication technologies (ICTs) as cognitive tools: Lessons learned from a design-based research study. *Computers & Education, 79*, 101–115.
Wang, Z., Pan, X., Miller, K. F., & Cortina, K. S. (2014b). Automatic classification of activities in classroom discourse. *Computers & Education, 78*, 115–123.
Wang, M., Chen, Z., Liu, Q., Peng, X., Long, T., & Shi, Y. (2025). Understanding teachers' willingness to use artificial intelligence-based teaching analysis system: extending TAM model with teaching efficacy, goal orientation, anxiety, and trust. *Interactive Learning Environments*, *33*(2), 1180–1197.
Wenger, E., McDermott, R., & Snyder, W. M. (2002). Seven principles for cultivating communities of practice. *Cultivating Communities of Practice: A Guide to Managing Knowledge*, *4*, 1–19.
Williams, M. D., Rana, N. P., & Dwivedi, Y. K. (2015). The unified theory of acceptance and use of technology (UTAUT): A literature review. *Journal of Enterprise Information Management, 28*(3), 443–488. https://doi.org/10.1108/jeim-09-2014-0088
Wise, A. F., & Jung, Y. (2019). Teaching with analytics: Towards a situated model of instructional decision-making. *Journal of Learning Analytics, 6*(2), 53–69.
Yang, H. H., Feng, L., & MacLeod, J. (2019). Understanding college students' acceptance of cloud classrooms in flipped instruction: Integrating utaut and connected classroom climate. *Journal of Educational Computing Research, 56*(8), 1258–1276.
Yoon, H. Y., Kang, S., & Kim, S. (2024). A non-verbal teaching behaviour analysis for improving pointing out gestures: The case of asynchronous video lecture analysis using deep learning. *Journal of Computer Assisted Learning, 40*(3), 1006–1018.
Zalavra, E., Papanikolaou, K., Dimitriadis, Y., & Sgouropoulou, C. (2023). Representing learning designs in a design support tool. *Education and Information Technologies, 28*(6), 6563–6594.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.